WEEK 1 WORK REPORT

Cybersecurity Lab and Foundational Skills

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Week	Week 1	
Work period	17 OCT – 23 OCT	
Prepared on	23 OCT	

1. Purpose and Scope

Purpose: Establish an isolated virtual cybersecurity practice environment and build entry-level skills in Linux, networking, and cybersecurity.

Activities: VirtualBox setup, network configuration, Linux account and permission exercises, network diagnostics, Try-HackMe learning, OverTheWire Bandit, and documentation.

2. Virtual Lab Inventory

Virtualization platform: VMware

VM: Kali Linux, Ubuntu, Windows 10

Virtual network: Host-only or NAT Network

Evidence: Store VM inventory, adapter configuration, IP settings, and snapshot screenshots in Findings/Lab Setup/.\

3. Connectivity and Network Checks

Addressing: Record each VM's interface address, prefix/subnet mask, gateway, and DNS server from actual command output.

Tests performed: Use ping for permitted connectivity checks, and traceroute/tracepath and nslookup/dig for diagnostics where available.

Observations: Record successful and failed tests, likely causes, and troubleshooting. Firewalls may block ICMP or traceroute probes.

Network diagram: Create a draw.io diagram based on the actual configuration and save the editable file plus PNG/PDF export in Findings/Network Map/.

4. Ubuntu Administration Tasks

User accounts: Three accounts: [identifiers].

Groups: Two groups: [group names].

Shared directory: Path: [path] | Owner/group: [owner:group] | Permissions: [mode] | Access validation: [actual result].

Command practice: Document at least 30 commands with purpose and actual output, including find, grep, ps, and ss -tulpn.

Scheduled task: Cron purpose: [task] | Schedule: [expression] | Verification: [actual result].

5. Guided Cybersecurity Practice

TryHackMe: Keep an individual writeup for each assigned room: Search Skills; Learning Cyber Security; Linux Fundamentals Part 1; What is Networking; Introductory Networking; Networking Concepts; Pyramid Of Pain; Intro to LAN.

OverTheWire: Track Bandit levels 0–10 separately, describing approach, useful commands, mistakes, and learning points.

6. Knowledge Summary

Security fundamentals: Confidentiality limits information access to authorized parties; integrity protects accuracy and authorized changes; availability keeps systems usable when needed.

Operational roles: SOC analysts monitor and triage alerts; penetration testers identify weaknesses within an agreed scope; incident responders investigate, contain, and support recovery.
Network foundations: IP supports addressing and routing, MAC addresses support local-link delivery, DNS resolves names, DHCP leases network settings, and ports identify service endpoints.

Linux skills: Practice directory navigation, file searching, permissions, ownership, process inspection, logs, and scheduled jobs.

Risk vocabulary: A threat may cause harm; a vulnerability is a weakness; an exploit uses a weakness; risk considers likelihood and impact.

7. Safe and Ethical Working

Permission: Test only devices, networks, and applications owned by you or explicitly authorized for testing.

Isolation: Keep deliberately vulnerable VMs on Host-only or a carefully configured NAT Network. Avoid Bridged mode for vulnerable targets.

Evidence handling: Use genuine screenshots, avoid unnecessary sensitive data collection, and redact unrelated personal information.

Legal awareness: For work performed in India, the Information Technology Act, 2000, as amended, may be relevant. This is not legal advice; check current official law.

Evidence Checklist

Area	Evidence to attach
VM setup	VM list, adapter settings, IP configuration, clean snapshots
Linux exercises	Accounts, groups, shared-folder permissions, 30 command outputs, cron verification
Networking	IP/gateway/DNS inventory, connectivity results, draw.io source and export
Training rooms	Individual TryHackMe writeups and completion evidence
Bandit	Level 0–10 writeups and screenshots with passwords hidden
Final documents	Cheat sheet, PDF report, presentation, README